

Response to Reviewers — Round 2

Manuscript: Spatial Cluster Randomized Trials — Sampling Design with Spillover Effects & Spatial Dependence

Journal: BMC Medical Research Methodology

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Revised Manuscript File: Revisions_V2.tex

Dear Editors (Michael Grayling & Vanshali Mota) and Reviewer,

We thank the Editor and the Reviewer for their continued thorough and constructive engagement with our manuscript. We have carefully considered each of the six major concerns raised and have revised the manuscript accordingly. Below we provide a detailed point-by-point response to each comment. All changes are implemented in the revised file Revisions_V2.tex. Page and line numbers refer to the revised manuscript.

The revised manuscript preserves the core scientific contribution---comparing Simple Random Sampling and Block Stratified Sampling for spatial CRT design---while substantially improving the precision and clarity of the model specification, notation, and inferential framing in response to the Reviewer's concerns.

Point-by-Point Responses

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
C1	The parameter beta is described as the 'total treatment effect' in Section 2.3.1. In the SAR model, beta represents only the direct treatment effect; the indirect spillover effect is captured by psi, and the true total effect involves both components. This mislabeling is conceptually important.	Section 2.3.1 (intervention-effect paragraph, ~line 283) and new "Estimands and Interpretation" paragraph in the SAR model section (~line 535), Revisions_V2.tex

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We thank the Reviewer, who is correct on the mathematics, and we have substantially revised our treatment of the estimands rather than restating the earlier position. We now state explicitly in a new "Estimands and Interpretation" paragraph (Section 2, SAR model) that β, ψ, and ρ are structural coefficients of the spatial autoregressive model, not marginal potential-outcome contrasts. Because the SAR equilibrium (reduced) form is $y = (I - \rho W)^{-1}(\alpha 1 + X\beta + Z\psi + \epsilon)$, a change in any cluster's treatment status propagates through the spatial multiplier $(I - \rho W)^{-1}$; consequently a potential-outcome indirect effect such as $E[Y(0,1)] - E[Y(0,0)]$ is a derived quantity that depends jointly on β, ψ, ρ, and W and equals no single coefficient in the model---exactly as the Reviewer observes---and with ρ not equal to 0 even "control-only" spillover induces a treated-to-treated mechanism. We have therefore removed the incorrect statement that the total treatment effect equals $\hat{\beta} + \hat{\psi}$: that sum ignores the autoregressive multiplier and is not the potential-outcome contrast. The parameter β is now defined consistently as the direct structural coefficient, and we clarify that omitting z_i from a fitted model causes the coefficient on the treatment indicator to conflate the direct and indirect effects through omitted-variable bias (made quantitative in our response to Comment 4). The loose "beta-hat ... total treatment effect" sentence in Section 2.3.1 has been deleted.	

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C2	Spillover is described as $\psi(d_i)$, a function of distance from the 'nearest intervention source,' but (a) 'intervention source' is never defined (cluster boundary? treated unit? individual?), (b) the model later treats spillover as a binary Rook-neighbor indicator creating inconsistency, and (c) ψ is used as a scalar in model (6). Individual-level locations are simulated but may be discarded if spillover operates only at the cluster level.	SAR model Eq. (3) description (~line 529); exact-collinearity / identifiability statement in the estimation-model paragraph (~line 623); subject-coordinates note in simulation procedure (~line 741)

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We agree that the $\psi(d_i)$ notation was misleading and have corrected it throughout. The simulation uses exclusively binary spillover based on Rook contiguity: $z_i = 1$ if cluster i shares a Rook edge with at least one treated cluster (a cluster assigned to receive the intervention), and 0 otherwise. The distance-function notation $\psi(d_i)$ has been replaced with the scalar ψ; 'intervention source' is now explicitly defined as 'treated cluster'; and z_i is defined as a binary Rook-neighbor indicator. Individual-level observations within each cluster share the cluster-level treatment status x_i and spillover status z_i---these are constant for all subjects k within cluster i. We retain Appendix D for the distance-based neighbor extension to clarify it is a generalisation, not the simulation's implementation. Most importantly, the Reviewer is correct that for many small-grid allocations the treatment and spillover indicators are collinear, so that ψ is not separately identifiable; we now address this head-on rather than leaving it implicit. Under block stratified sampling with control-only spillover the checkerboard forces $z_i = 1 - x_i$ exactly, so β and ψ cannot be separately estimated and $\hat{\beta}$ estimates $\beta - \psi$ (bias = $-\psi$). This identifiability statement has been added to the estimation-model paragraph (Section 3.1), and we show it matches the observed BSS/TrtNoSpill bias in Table 1 (see our response to Comment 4). Finally, we confirm that the individual within-cluster coordinates illustrated in Figure 4 do not enter the SAR fit---the	

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C3	Multiple notation issues: w_{ij} is not defined at Equation (1); y is indexed by i and k but x and ϵ are not; Equations (4)–(6) drop the cluster index without explanation; the W matrix definition is ambiguous regarding whether it is cluster-level or individual-level.	After Eq. (1) (~line 520) and model description (~lines 525–529); Appendix A (w_{ij}/W_{ij} row-normalization and cluster-level wording)

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We have addressed all seven sub-items systematically. (3a) w_{ij} undefined in Equation (1). An inline definition has been added immediately after Equation (1): w_{ij} are elements of the row-normalized spatial weights matrix W (equal to $1/ N(i)$ if clusters i and j share a Rook edge, and 0 otherwise; full construction in Appendix A). (3b) Inconsistent indexing: y indexed by both i and k in Eq (1) but x and ϵ are not; x and z have i,k indexes in Eq (3). We have harmonized the indexing throughout. In Equation (1), ϵ_i has been corrected to ϵ_{ik}, consistent with y_{ik}. In Equation (3), x_{ik} and z_{ik} have been corrected to x_i and z_i, reflecting that treatment and spillover status are cluster-level quantities shared by all subjects k within cluster i. The error term ϵ_i in Eq (3) has also been corrected to ϵ_{ik}. The notation now consistently distinguishes cluster-level covariates (subscript i only) from individual-level outcomes and errors (subscript ik). (3c) Cluster index dropped in Equations (4)–(6). The ik indexing is now carried through all equations, including the spatial dependence term (Eq 4), the estimation equation (Eq 5), the data-generating expression, and the Appendix B likelihood expressions. This makes the multi-level structure explicit and consistent throughout. Within each cluster, because x_i and z_i are cluster-level constants, individual responses y_{ik} differ only through the error term ϵ_{ik}. (3d) W matrix level ambiguity. W is now described explicitly as a cluster-level $n \times n$ adjacency	

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C4	The paper describes the data-generating process (DGP) but never explicitly states which model is fitted in the simulation. If the spillover indicator z_i is omitted from the estimation model this must be stated; if included, the near-collinearity between x_i and z_i needs to be addressed.	Section 3.1, estimation-model paragraph following Eq. (EstModel), with exact-collinearity derivation and Table 1 cross-reference (~lines 616--635), Revisions_V2.tex

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	The Reviewer has reverse-engineered the mechanism correctly, and we now state it explicitly. We added a new paragraph immediately after the DGP equation (Equation DataGen) giving the estimation model (new Eq. EstModel), which has the same structural form as the DGP: $y_i = \rho \sum_{j \neq i} w_{ij} y_j + \alpha + \beta x_i + \psi z_i + \epsilon_i$, estimated via <code>lagsarlm()</code> MLE. The fitted model therefore does include the spillover term ψz_i; bias in $\hat{\beta}$ does not arise from deliberately omitting a variable. Rather, it arises because for an entire family of allocations the collinearity between x_i and z_i is exact, not merely high. Under block stratified sampling with control-only spillover (<code>TrtNoSpill</code>) the checkerboard yields $z_i = 1 - x_i$ for all clusters, so the design columns for x_i and z_i are perfectly collinear and β, ψ are not separately identifiable. Writing $\alpha + \beta x_i + \psi z_i = (\alpha + \psi) + (\beta - \psi) x_i$ shows the intercept absorbs ψ and $\hat{\beta}$ estimates $\beta - \psi$, giving bias = $-\psi$---which is exactly what the Reviewer anticipated as omitted-variable bias. This prediction is confirmed by the authors' own results: for BSS/<code>TrtNoSpill</code> at $\rho = 0$ the estimated bias of $\hat{\beta}$ is -0.496, -0.597, -0.697, -0.798 at $\psi = 0.5, 0.6, 0.7, 0.8$ (Table 1), matching $-\psi$ to within Monte Carlo error. We have added this derivation and the Table 1 cross-reference to Section 3.1, and the Limitations section discusses the identifiability challenge.	

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
C5	BSS has only 2--6 valid allocations (nearly deterministic). The response from Revision 1 states 'BSS is a design-stage tool evaluated through exhaustive enumeration, not a randomization distribution for permutation inference,' but this does not fully address the frequentist inference problem. If treatment allocation is nearly deterministic, the permutation-based randomization test for BSS would have almost no power. What practical value does BSS have if it cannot be used for inference?	Section 3.1 inference paragraph expanded (~line 679); Limitations paragraph on near-deterministic allocation and model-based inference (Section 5.2, ~line 1238)

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We appreciate this important challenge and have expanded our treatment of it. The key distinction is between design selection (using BSS vs. SRS to choose an allocation) and inference (how to construct confidence intervals and p-values). We have expanded the design-stage inference paragraph in Section 3.1 to state explicitly: BSS is a design-selection criterion for identifying a spatially favorable allocation before data collection. Because the valid BSS allocation space is near-deterministic (2--6 arrangements), permutation-based randomization tests under BSS are infeasible. Investigators using BSS should conduct inference via model-based SAR maximum likelihood estimation, not permutation tests. This is now also stated as an explicit limitation in Section 5.2 (Limitations). The practical value of BSS lies in pre-study design evaluation: before beginning the trial, investigators can compare BSS vs. SRS in terms of expected worst-case estimation error under their anticipated spillover regime and grid geometry---as this simulation study demonstrates. We acknowledge that this repositions BSS: it is a constrained design whose justification rests on the bias-variance properties we characterise and which must be paired with model-based inference, never with permutation inference. We have stated this limitation plainly rather than claiming a randomization-inference role for BSS.	

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C6	According to the results, the bias from these designs can be large. The reviewer argues that 'fried egg' or buffer zone designs should be recommended over BSS, especially under TrtSpill. The results suggest that the proposed designs may not be suitable for practical use.	Discussion severity paragraph (~line 1228); Future Considerations buffer-zone paragraph (~line 1285); conditional-recommendation edits in Abstract, Application, and Conclusion, Revisions_V2.tex

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We acknowledge the Reviewer's concern and have made two changes: we quantify "severe" and we temper the recommendation. First, we take up the Reviewer's marginal query---"what is severe?"---directly. With a true effect $\beta = 1.0$ and spillover ψ in $[0.5, 0.8]$, the bias of $\hat{\beta}$ under exactly collinear allocations equals $-\psi$, i.e. 50-80% of the true effect, large enough to reverse a practical conclusion; we now state severity in these effect-size units in the Discussion and agree the error can indeed be severe. Crucially, the severity is conditional, not universal: it is largest precisely where identifiability is lost (BSS under TrtNoSpill, and the checkerboard under $\text{TrtSpill} + \rho > 0$) and shrinks toward a few percent of β where the allocation lets ψ be separately estimated (many SRS allocations under TrtSpill; Table 1). Second, we have softened the manuscript's previously broad endorsement of BSS---in the Abstract, Discussion, Application, and Conclusion---to an explicitly conditional recommendation. We have also added buffer-zone ("fried-egg") designs as an important future direction (Section 5.3), and we note there that excluding boundary control clusters restores $z_i = 0$ for the analyzed controls, breaking the exact $x_i - z_i$ collinearity by design and thereby addressing the root cause of the large biases rather than only their variance. We respectfully retain the paper's contribution on the following grounds: (1) the primary contribution is characterising the bias-variance and identifiability landscape across grid geometries and spillover regimes; knowing that	

Summary of All Changes

The table below lists all changes made in Revisions_V2.tex for this revision round.

ID	Change Summary	Status
C1	Section 2.3.1 (intervention-effect paragraph, ~line 283) and new "Estimands and Interpretation" paragraph in the SAR model section (~line 535), Revisions_V2.tex	Done
C2	SAR model Eq. (3) description (~line 529); exact-collinearity / identifiability statement in the estimation-model paragraph (~line 623); subject-coordinates note in simulation procedure (~line 741)	Done
C3	After Eq. (1) (~line 520) and model description (~lines 525--529); Appendix A (w_ij/W_ij row-normalization and cluster-level wording)	Done
C4	Section 3.1, estimation-model paragraph following Eq. (EstModel), with exact-collinearity derivation and Table 1 cross-reference (~lines 616--635), Revisions_V2.tex	Done
C5	Section 3.1 inference paragraph expanded (~line 679); Limitations paragraph on near-deterministic allocation and model-based inference (Section 5.2, ~line 1238)	Done
C6	Discussion severity paragraph (~line 1228); Future Considerations buffer-zone paragraph (~line 1285); conditional-recommendation edits in Abstract, Application, and Conclusion, Revisions_V2.tex	Done

Closing

We believe the revised manuscript fully addresses the Reviewer's concerns. We have clarified the estimand definitions, corrected the spillover notation, resolved indexing inconsistencies, explicitly stated the DGP and estimation model, strengthened the inferential framing of BSS, and added buffer-zone designs as a future direction. We hope that these revisions demonstrate the methodological rigor and practical value of our work, and we look forward to hearing from you.

We remain grateful for the Reviewer's expertise and commitment to improving this manuscript.

Sincerely,

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